ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF ENVIRONMENTAL AND EARTH SCIENCES

(RSEES)

DEPARTMENT OF GEOGRAPHY

M.Sc. in Geoinformatics

Postgraduate Programme as per NEP, 2020

W.E.F

AY-2025-26

M.Sc. in Geoinformatics

Programme Structure

1st SEMESTER				
COURSE CODE	**COURSE TITLE**	**LEVEL**	**CREDIT**	**L-T-P**
GEOI164C101	Principles of Remote Sensing; Global Positioning System	500	4	3-1-0
GEOI164C102	Fundamentals of GIS	400	4	3-1-0
GEOI164C103	Cartography & Geo Statistics	400	4	3-1-0
GEOI164C104	Geosciences & Image Interpretation	400	4	3-1-0
GEOI164C115	Techniques in Spatial Data Processing and Cartography	400	4	0-0-8
	Swayam Course (Upcoming)		3/4	
	TOTAL CREDIT FOR 1st SEMESTER		**20+ 3/4**	
2nd SEMESTER				
COURSE CODE	**COURSE TITLE**	**LEVEL**	**CREDIT**	**L-T-P**
GEOI164C201	Digital Image Processing	500	4	3-1-0
GEOI164C202	Spatial Analysis & Modelling	500	4	3-1-0
GEOI164C203	Geoinformatics in Agriculture, Soil & Land Evaluation	500	4	3-1-0
GEOI164C204	Geoinformatics in Regional and Urban Planning	500	4	3-1-0
GEOI164C215	Remote Sensing and Digital Image Processing and Analysis	400	4	0-0-8
	Swayam Course (Upcoming)		3/4	
	TOTAL CREDIT FOR 2nd SEMESTER		**20+3/4**	
TOTAL CREDIT FOR 1st YEAR = 46/48				
3rd SEMESTER				
COURSE CODE	**COURSE TITLE**	**LEVEL**	**CREDIT**	**L-T-P**
GEOI164C301	Advanced Remote Sensing	500	4	3-1-0
GEOI164C302	Research Methodology	500	4	3-1-0
GEOI164C313	Techniques in Advanced Remote Sensing and Spatial Modelling	500	4	0-0-8
GEOI164D304	Geoinformatics in Disaster Management	500	4	3-1-0
GEOI164D305	Geoinformatics in Forestry	500	4	3-1-0
OR 3rd SEMESTER (For students with 3rd and 4th Semester Research)				
GEOI164C321	**RESEARCH PROJECT – PHASE I**	500	20	
Coursework (12 credits) + Research work (8 credits)				
GEOI164C301	Advanced Remote Sensing	500	4	3-1-0
GEOI164C302	Research Methodology	500	4	3-1-0
GEOI164C313	Techniques in Advanced Remote Sensing and Spatial Modelling	500	4	0-0-8
GEOI164C324	Research 1	500	8	
	TOTAL CREDIT FOR 3rd SEMESTER		**20**	

4th SEMESTER

COURSE CODE	COURSE TITLE	LEVEL	CREDIT	L-T-P
GEOI164C401	Geoinformatics in Geomorphological studies	500	4	3-1-0
GEOI164C402	Geoinformatics in Water Resources	500	4	3-1-0
GEOI164C403	Application of Geoinformatics in Climate Change studies	500	4	3-1-0
GEOI164C404	Geoinformatics in Land Resource Management	500	4	3-1-0
GEOI164C405	Geoinformatics in Environmental Management	500	4	3-1-0
OR 4th SEMESTER (For students with 3rd and 4th Semester Research)				
GEOI164D421	**RESEARCH PROJECT – PHASE II**	500	20	
Dissertation only (for students who have completed only coursework of 20-credit coursework in 3rd semester for Coursework only in lieu of research)				
GEOI164C401	Geoinformatics in Geomorphological study	500	4	3-1-0
GEOI164C402	Geoinformatics in Water Resources	500	4	3-1-0
GEOI164C423	Research (Major)	500	12	
TOTAL CREDIT FOR 4th SEMESTER= 20				
TOTAL CREDIT FOR 2nd YEAR = 40				

SEMESTER 1

Paper I Core Course	PRINCIPLES OF REMOTE SENSING & GLOBAL POSITIONING SYSTEM L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 101

Course Objectives: *The course contributes significantly toward sustainable technological advancement and spatially informed decision-making by integrating Remote Sensing, GIS, and GPS applications. It equips students with technical and analytical skills essential for Smart infrastructure planning, Sustainable urban development, Environmental and natural resource management etc. Thus, the course meaningfully supports the broader objectives of SDG 9 and SDG 11 through geospatial technology-driven sustainable development practices.*

Course Outcomes: After successful completion of course, the students will be able to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	**Define** the basic concepts and significance of Remote Sensing, GIS, and GPS technologies in sustainable development and infrastructure planning	**BT1**
CO2	**Explain** the principles, techniques, and applications of Remote Sensing and aerial photography for environmental and urban studies	**BT2**
CO3	**Apply** Remote Sensing data and geospatial technologies for natural resource assessment, disaster management, and sustainable urban planning	**BT3**
CO4	**Analyze** and utilize geospatial datasets from different platforms for infrastructure development and spatial decision-making	**BT4**
CO5	**Interpret** and evaluate geospatial data using GIS, Remote Sensing, and GPS techniques for geographical research and sustainable community planning	**BT5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Introduction to Remote Sensing: Concepts, definition, history, development; Physics of Remote Sensing: Electromagnetic Radiation (EMR), Theories of EMR, Laws of Radiation, EM Spectrum, Sources of EMR; Interaction of EMR: Interaction between radiation and matter, Interaction with Earth‟s Atmosphere, Atmospheric Window, Reflection, Absorption, and Transmission.	10
Unit 2	Spectral Signature: Spectral Signatures for common features, e.g. snow, soil, water and vegetation; platform and sensors: platforms, sensors, orbits: types of platform, types of sensors- Active and Passive, cameras and satellite orbits, concept of resolution, Satellite imaging modes.	10
Unit 3	Fundamentals of Radiometry: concept of solid angle, radiometric measurements, observation geometry in RS; data products and RS data errors: Satellite data generation, data reception, Type of data products and Aerial Photography products, FCC and TCC images and their applications, radiometric, geometric and atmospheric errors.	10
Unit 4	Photogrammetry: basics of Aerial Photography, basic geometry of aerial photograph, central and orthographic projections, difference between map and aerial photograph, types of aerial photographs. Measurements: scale and ground coverage of aerial photographs, Geometry of Aerial Photographs, determination of scale, use of Parallax, height measurement. Aerial Photo and image Interpretation: Elements of visual interpretation for aerial photos and satellite imageries: Single, vertical Stereo Pairs, derived from PAN, LISS, Wifs, OCM Sensors. study and visual Interpretation of Satellite Images for physical features, urban, forest and agricultural uses. Digital Photogrammetry: concept and techniques of digital photogrammetry.	10
	Total	**40**

Text Books:

1. Jensen, J.R., (2006) "Remote Sensing of the Environment – An Earth Resources Perspective", Pearson Education, Inc. (Singapore) Pte. Ltd., Indian edition, Delhi.
2. George Joseph, (2004) "Fundamentals of remote sensing", Universities press (India) Pte Ltd., Hyderabad.

Reference books:

1. Sabins, F.F. Jr., (2007) Edition. „Remote Sensing – Principles and Interpretation", W.H. Freeman & Co.
2. Reeves, Robert G. (1991), "Manual of Remote Sensing, Vol. I, American Society of Photogrammetry and Remote Sensing, Falls Church, Virginia, USA
3. Lillesand, Thomas M. and Kiefer, Ralph, W., (2007) "Remote Sensing and Image Interpretation", 4th Edition, John Wiley and Sons, New York
4. Rampal, K.K., (1999) Handbook of Aerial Photography and Interpretation, Concept Publishing Company, New Delhi
5. N.K.Agrawal , (2004) ,Essentials of GPS, Spatial Network Pvt. Ltd
6. Sathish Gopi, (2000), GPS and Surveying using GPS
7. Leica. A., (2003), GPS Satellite Surveying, John Wiley & Sons, use. New York
8. Terry-Karen Steede, (2002), Integrating GIS and the Global Positioning System, ESRI Press

Paper II Core Course	FUNDAMENTALS OF GIS L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 102

Course Objectives: *This course aims to develop students' understanding of Geoinformatics technologies including Remote Sensing, GIS, and GPS for innovative spatial data analysis, infrastructure planning, mapping, and sustainable decision-making applications supporting industrial and technological development. This course aligns with SDG 9.*

Course Outcomes:

After the completion of the course, the students will have the ability to:

Detailed Syllabus:

Course Outcome CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Recall fundamental GIS concepts, terminologies, and geospatial technologies used in infrastructure and development planning	**BT1**
CO2	Construct raster and vector maps using GIS techniques for spatial representation and infrastructure analysis	**BT2**
CO3	Develop thematic maps and geospatial databases for industrial planning, resource management, and spatial decision-making	**BT3**
CO4	Analyze GIS and Remote Sensing data to perform spatial analysis, infrastructure assessment, and GPS-based mapping applications	**BT4**
CO5	Assess and apply GIS techniques and geospatial technologies in innovation-driven sectors including urban infrastructure, transportation, environmental management, and regional planning.	**BT5**

Modules	Topics and Course Content	Periods
Unit 1	Basic concepts of GIS: Definition and history; Components of GIS; Data structure and formats; Spatial data models – Raster and Vector; Data base design - editing and topology creation in GIS, Linkage between spatial and non-spatial data; Data inputting in GIS	12
Unit 2	Integration of Raster & Vector Data; Cartographic Modeling - Map Algebra; Raster Data & its Representation: Types, Data Structure, Data Compression, Data Files, Data Conversions; Raster Data Analysis – Overlay Operations, Slope & Aspects, Statistical Analysis; Geometric Transformations - Affine Transformation and Geometric Transformation Coefficients, RMS Error; Vector data representation: Topological & Non-topological Vector Data, Map scale, Spatial Resolution, Spatial Data Accuracy, Location Data Accuracy and Precision, Vector Data Sources; Comparison between Raster & Vector Data; Feature Based Topological functions: Buffering Overlay Analysis, Distance Measurements; Layer Based Topological Functions	14
Unit 3	Vector Data Query, Attribute Data Query; Logical Expressions, Types of Operations; Relational Database Query: Use of SQL, Descriptive Statistics of Attribute Data; Spatial Data Query, Raster Data Query, Query by Cell Value, Query using Graphical Methods, Charts; Geographic Visualization, Data Classification, Spatial Aggregation, Map Comparison; Problem Identification & Designing a Data Model	10
Unit 4	Application of GIS Techniques in various fields; Web GIS	04
	Total	**40**

Text Books:

1. Burrough, Peter A. and Rachael McDonnell,(1998), „ Principles of Geographical Information Systems" Oxford University Press, New York.
2. C.P.L and Albert K.W.Yeung (2006) "Concepts and Techniques of Geographic Information Systems" Prentice Hall of India,New Delhi.

Reference Books:

1. Demers, Michael N. 2000. *Fundamentals of Geographic Information Systems*. John Wiley, Singapore.
2. ESRI 1993. *Understanding GIS.* Redlands, USA
3. George, Joseph 2003. *Fundamentals of Remote Sensing*. Universities Press (Pvt.) Ltd, Hyderabad.
4. Girard, M-C. and Girard, C. M. 2003. *Processing of Remote Sensing Data*. Oxford & IBH, New Delhi.

5. Heywood, Ian 2003. *An Introduction to Geographical Information Systems*. 2nd ed. Pearson Publ. Co., Singapore.
6. Kang-tsung Chang (2007),,,Introduction to Geographic Information Systems" Tata McGraw Hill, New Delhi.
7. Longley, P., Goodchild, M.F., Maguire, D. and Rhind, D. 1999. *Geographic Information Systems. Principles, Techniques, Management, Applications*. John Wiley, New York.
8. Magwire, D. J., Goodchild, M.F. and Rhind, D. M., (2005),,,Geographical Information Systems: Principles and Applications', Longman Group, U.K.
9. Martin, D. 1996. *Geographic Information Systems: Socioeconomic Implications*. Routledge, London.
10. Ralston, B. A. 2002. *Developing GIS Solutions with Map Objects and Visual Basic*. OnWord Press: Thompson Learning, New York & Singapore.
11. Reddy, M. Anji 2001. *Textbook of Remote Sensing and Geographic Information Systems*. B. S. Publs., Hyderbad.

Paper III Core Course	CARTOGRAPHY & GEOSTATISTICS L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 103

Course Objectives: *This course aims to develop foundational and applied knowledge in Cartography and Geostatistics through modern geospatial technologies, enhancing analytical, technical, and problem-solving skills for quality education, innovation, and infrastructure-related spatial applications. This course aligns with and SDG 4 and SDG 9.*

Course Outcomes:

After successful completion of the course, the students will be able to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Define fundamental concepts of cartography, map scales, coordinate systems, and topographic maps for geospatial learning and professional applications	**BT1**
CO2	Explain the principles and techniques of cartographic design for effective spatial communication and knowledge development	**BT2**
CO3	Construct digital cartographic maps and spatial databases using modern geospatial data structures and tools	**BT3**
CO4	Analyze spatial databases and perform geostatistical queries for infrastructure planning and data-driven decision-making	**BT4**
CO5	Interpret and evaluate geostatistical analysis results in GIS platforms for innovative spatial problem-solving and sustainable development applications	**BT5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Introduction to cartography: nature and scope; Approximation of Earth, map projection and coordinate system: concepts, types and uses; categories & characteristics of maps, study of different types of maps, Survey of India national series maps, interpretation of topographic maps; basics of map scales; indexing and numbering of topographical maps	12
Unit 2	Fundamentals of cartographic design, colour, pattern, lettering, compilation, border information, aesthetics; Generalization: semantic & geometric, symbolization, dot, isopleth and choropleth mapping; multivariate and dynamic mapping; map production, methods of map printing; visualization of geospatial data: design aspects, multiscale and geometric aspects scale, dissemination of (visualized) geospatial data, graphic symbology & variables; data products, use and users of products	12
Unit 3	Digital Cartography - elements of digital Cartography; analog to digital conversion of data; conventional mapping v/s digital mapping; nature of data, database and data structures; data input: data capture, digitization and scanning; digital database creation: point features, line features, polygon features; data editing-removal of errors – overshoot & undershoot, snapping; data collection and integration, non-spatial data attachment working with tables; dissolving and merging	12

Unit 4	Data base query: Reclassification, overlay cross tabulation, editing, assigning attribute values, extraction of attribute values, histogram, area and perimeter calculation, profile generation, probability classification; Mathematical operations: Image overlay, scalar image operations, image attribute transformation; Distance operators: Distance analysis (spherical distance, cost distance), buffer analysis, direction variable cost distance, dispersion distance, least cost path analysis, spatial allocation and reallocation, Thiessen Polygon; Context operators: Surface analysis, filtering pattern analysis, grouping watershed, determination, hinterland determination; Statistics: Regression analysis (multiple, logistic, pattern analysis, trend surface analysis, spatial auto correlation, quadrant analysis, weighted mean, centre/ standard radius, compaction index, sampling (random, systematic and stratified), standard scores method.	12
	Total	**48**

Text Books:

1. Keates, J.S., (2008): Cartographic Design and production, London, Longman
2. Ramesh, P. A., (2000): Fundamentals of Cartography, Concept Publishing Co., New Delhi.

Reference Books:

1. Rampal, K.K., (2004): Mapping and Compilation, Concept Publishing Co.,New Delhi.
2. Anson, R.W.& Ormeling, F.J., (2008), Basic Cartography, Vol. I&II ed., Elsevier Applied Science Publishers, London.
3. Robinson A.H. & Morrison J.L, (1995) Elements of Cartography, John Wiley & Sons
4. Singh, R.L & Dutt. P.K, (2008),"Elements of Practical geography", Students Friends Allahabad
5. Peterson, M.P., (1995) "Interactive and Animated Cartography" Upper Sadde River, NJ: Prentice Hall.
6. Clark, I. (1979), Practical Geostatistics, Applied Science Publishers, London
7. Davis, J.C. (1973), Statistics and Data Analysis in Geology, Wiley, New York.
8. Matheron, G.F, (1963) Principles of Geostatistics: Economic Geology vol.58
9. Stein, A. (1998), Spatial Statistics for Soils and the Environment, ITC lecture notes.

Paper IV Core Course	GEO SCIENCES & IMAGE INTERPRETATION L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 104

Course Objectives: *This course aims to develop an understanding of Geosciences and image interpretation techniques for environmental monitoring, climate change assessment, natural resource management, and sustainable land ecosystem analysis using geospatial technologies. This course is aligned with SDG 13 and SDG 15.*

Learning Outcomes:
After the completion of the course, the students will have the ability to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Define fundamental concepts of Geosciences, image interpretation, and topographic analysis related to environmental and land resource studies	**BT1**
CO2	Explain the principles and techniques of image interpretation for monitoring climate and terrestrial ecosystems	**BT2**
CO3	Apply geospatial and image interpretation techniques for mapping landforms, vegetation, natural resources, and environmental changes	**BT3**
CO4	Analyze geoscientific and remotely sensed data for climate action, land degradation assessment, and ecosystem management	**BT4**
CO5	Evaluate and interpret geospatial analysis results for sustainable land resource planning, biodiversity conservation, and environmental management	**BT5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Concept of Earth System, Lithosphere, Biosphere, Hydrosphere & Atmosphere; Elements of Photo Interpretation in Geological Studies- lithotypes and structural features	10
Unit 2	Visual and Digital Satellite Image Interpretation; Elements of Image Interpretation; Development of Interpretation Keys; Ground Truth Verification	10
Unit 3	Fundamental Concepts: Geomorphic Agents and Processes; Development of Drainage Patterns and their Significance; Image Characteristics of Major Landforms: Fluvial, Aeolian, Glacial and Marine	10
Unit 4	Natural Hazard Risk Management; Regional & Urban Planning; Agricultural, Soil and Land Evaluation; Water Resources	10
	Total	**40**

Text Books:

1. Murk & Skinner, (1999). Geology Today – Understanding Our Planet, John Wiley And Sons Inc, New York.
2. Lillisand, T. M. and Keifer, R. W., (2007). Remote Sensing and Image Interpretation", John Willey and Sons, New York, Fourth Edition

Reference books:

1. Pandey, S. N. , (1987). Principles and Applications of Photogeology. New Delhi: Eastern Wiley.
2. Jenson, J.R., (2006). Remote Sensing of the Environment – An Earth Resource Perspective, Prentice Hall Inc.
3. Drury, S.A. , (2004). Image Interpretation in Geology, Chapman & Hall, India.
4. Thornbury, W. D., (I969): Principles of Geomorphology, John Wiley and Sons, New York
5. Sabins, Floyd F., (2007). Remote Sensing: Principles and Interpretation, 2nd Ed., Freeman, New York.

Paper V Core Course	Techniques in Spatial Data Processing and Cartography L-T-P-C: 0-0-8-4 Credit Units: 4 Scheme of Evaluation: (P)	Subject Code: GEOI164C115

Course Objectives: *This course aims to enhance students' practical knowledge and technical skills in GIS, Remote Sensing, and spatial data processing for innovative geospatial applications, infrastructure planning, resource management, and sustainable development studies. This course aligns with SDG 9.*

Course Outcomes:

By the end of this course the students will be able to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Define the principles and practical concepts of GIS, Remote Sensing, and spatial data processing technologies	**BT1**
CO2	Classify the nature, characteristics, and sources of geospatial data used in infrastructure and spatial planning applications	**BT2**
CO3	Develop technical skills in geospatial data processing, digital cartography, and spatial analysis using GIS and Remote Sensing tools	**BT3**
CO4	Apply geospatial techniques to analyze geographical problems related to geomorphology, climatology, population studies, and regional development.	**BT4**
CO5	Evaluate and utilize geospatial tools and technologies for infrastructure planning, natural resource management, and innovation-driven spatial decision-making.	**BT5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit I	**Lab Work:** Exercise 1: a) Data download (Toposheets, Open Street maps) b) Geo-referencing & Reprojection: image to image rectification, keyboard entry rectification - setting projection c) Raster Mosaicking and Clipping Exercise 2: Creating and Managing Vector Data: a) Adding vector layer b) Geodatabase Creation & digitizing entities like point, line and polygon data c) Vector Layer Formatting Exercise 3: Working with attributes and Data Organization (location, attributes, consistency, scale)	12
Unit 2	Exercise 4: Relational Data Base Query Exercise 5: a) Importing Spread sheets or CSV files b) Graphical Representation of Spatial data Exercise 6: Map algebra – raster processing tools	12
Unit 3	Exercise 7: a) Methods of data analysis: Buffer analysis b) surface interpolation, reclassification Exercise 8: Construction of Map scales: Simple, Comparative and Diagonal Exercise 9: Construction of different types of map projection; Conical projection, Cylindrical Projection, Zenithal Projection	12
Unit4	Exercise 10: Preparation of Base Map Exercise 11: Designing, Symbolization, Pattern and Shading techniques	4
		40

Books Recommended

1. Date, C.J., 1995 : An Introduction to Data Base System, 6th edition, Reading Messachusetts; Adderson Werley.
2. Fraser Taylor, D.R., (ed.), 1980 : Progress in Contemporary Cartography, John Wiley, Chichester U.K.
3. Fraser Taylor, D.R., (ed.), 1983 : Graphic Communication and Design in Contemporary Cartography, John Wiley & Sons Ltd. New York.
4. Jones, C., 1997 : Geographic Information Systems and Computer Cartography, Longman, London.
5. Kraak, M-J., and Ormeling, F., 2004: Cartography: Visualization of Geospatial Data, Pearson Education.
6. Misra, R.P., et al 2014: Fundamentals of Cartography, Concept Publishers, Delhi.

Semester-II

Paper I Core Course	DIGITAL IMAGE PROCESSING L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 201

Course Objectives: *This course aims to develop understanding of digital image processing systems and advanced geospatial technologies for resource analysis, infrastructure assessment, spatial innovation, and technical skill development in geospatial applications. This course aligns with SDG 9*

Course Outcomes:

After the completion of the course, the students will have the ability to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Recall the fundamental concepts of digital image processing and characteristics of digital images used in geospatial applications	**BT1**
CO2	Explain image enhancement and filtering techniques for improved spatial data interpretation and infrastructure analysis	**BT2**
CO3	Apply multi-band enhancement and classification techniques using geospatial technologies for resource and infrastructure mapping	**BT3**
CO4	Analyze pattern recognition methods and classification techniques for advanced geospatial analysis and spatial decision-making	**BT4**

Detailed Syllabus:

Modules	Topics And Course Content	Periods
Unit 1	Introduction: Concepts about digital image and its characteristics, spectral, spatial, radiometric and temporal resolution, visual vs. digital methods, image data storage and retrieval, image restoration and noise abatement, radiometric and geometric correction technique, interpolation methods – linear and nonlinear transformation for geometric corrections	14
Unit 2	Image Enhancement & Filtering Techniques: Look-Up Tables (LUT) and types of image displays and fcc; image enhancement techniques: radiometric and spatial; contrast stretching: linear and non-linear methods; spatial filtering: high and low frequency, image smoothing,	12
Unit 3	Multi-Band Enhancement Techniques & Classification: Band ratio, types of vegetation indices; Principal Component Analysis (PCA), multi dated data analysis and change detection; digital image classification: supervised & unsupervised, accuracy assessment, error matrix	12
Unit 4	Pattern Recognition: Concept of pattern recognition, multi-spectral pattern recognition; spectral discrimination, signature bank, parametric and non-parametric classifiers; Kriging	10
	Total	**48**

Text Books:

1. Sabins, Floyd F. (2007), Remote Sensing: Principles and Interpretation, H. Freeman and C., New York.
2. Thomas M. Lillesand & Kiefer, Ralph W. (2007), Remote Sensing and Image Interpretation, John Wiley & Sons, New York.

Reference books:

1. Jensen, JR. (2006), Remote Sensing of the Environment- An Earth Resources Perspective, Prentice Hall Inc.
2. Rencz, Andrew N. , (1999), Remote Sensing for the Earth Sciences: Manual of Remote Sensing, 3rd ed., John Wiley & Sons, Inc., New York.
3. Curran, P., (1985), Principles of Remote Sensing, Longman, London.
4. Campbell, James B., (2006), Introductory Remote Sensing: Principles and Concepts, Routledge.
5. Gibson, P.J., (2000), Introduction to Remote Sensing, 2nd ed., Taylor & Francis, London.
6. Cracknell, A.P. & Hayes, L.W B., (2007), Introduction to Remote Sensing, Taylor & Francis, London.

Paper II Core Course	SPATIAL ANALYSIS & MODELLING L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 202

Course Objectives: *This course aims to develop theoretical and practical understanding of GIS, GPS, spatial analysis, and modelling techniques for innovative geospatial applications, infrastructure planning, spatial decision-making, and sustainable development. This course aligns SDG 9 and SDG 11.*

Course Outcomes:

After successful completion of the course, the students will be able to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Define the fundamental concepts of GIS, spatial analysis, and geospatial modelling techniques	**BT1**
CO2	Explain and interpret spatial data analysis methods used in infrastructure and development planning	**BT2**
CO3	Apply geostatistical and spatial interpolation techniques for geospatial problem-solving and resource analysis.	**BT3**
CO4	Analyze and utilize spatial modelling and decision support system frameworks for infrastructure planning and spatial decision-making applications	**BT4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Introduction to GIS analysis & modelling: Spatial Data: definition, analysis, processes & steps, software and tools; Raster-based and Vector-based GIS modeling, Binary Models, Index Models, Regression Models, Process Models; Geodatabase Model, role of databases in GIS, Creating, editing and managing	12
Unit 2	Spatial data analysis techniques: Classification scheme of Vector-based and Raster-based GIS operations; Raster–Based techniques: methods of Reclassification, Overlay analysis, Slope and Aspect, Buffering, Cost-Distance Calculation; Vector-based techniques: map manipulation techniques, Buffering, Overlay analysis, Network analysis; Digital Terrain Analysis and modeling: TIN and DEM, surface representation & analysis	12
Unit 3	Geostatistical analysis techniques: Introduction to spatial interpolation: control points; global methods: Trend Surface Analysis, Regression Models; Local Methods: Thiessen Polygons, Density Estimation, Inverse Distance Weighted (IDW) Interpolation; Kriging: Ordinary Kriging (Semivariance, Semivariogram), Universal Kriging	12
Unit 4	INTRODUCTION TO DSS GIS and decision support system, Introduction to decision making process and decision support systems, Introduction of a framework for planning and decision making, Spatial Decision Making; development of DSS, DSS architecture; Principles and components of multiple-criteria decision making; Main multiple- criteria evaluation methods/techniques; Spatial multiple criteria decision making; Multiple-criteria decision making in spatial data analysis; Introduction to AHP, Basic Principles of AHP; Effect Table, Pair Wise comparison, Standardization, Consistency, Wieghtage, performance score, Different method in PWC	12
	Total	**48**

Text Books:

1. Bonczek, R.H., C.W. Holsapple, and A.B. Whinston, (1981), Foundations of Decision Support Systems, Academic Press, New York. Basic text on DSS
2. Geoffrion, A.M., (1983). "Can OR/MS evolve fast enough? Interfaces 13:10. Source for six essential characteristics of DSS

Reference Books:

1. House, W.C. (ed.), (1983). Decision Support Systems, Petrocelli, New York. Basic DSS text
2. Sprague, R.H., (1997). "A framework for the development of decision support systems, "Management Information Sciences Quarterly 4:1-26. Source for DSS development model
3. Sprague, R.H., and Carlson, E.D., (1982). Building Effective Decision Support Systems, Prentice-Hall, Englewood Cliffs NJ. Basic DSS text
4. Burrough, Peter A. and Rachael McDonnell., (1998), Principles of Geographical Information Systems. Oxford University Press, New York
5. Laurini, Robert and Derek Thompson. , (1992), Fundamentals of Spatial Information Systems. Academic Pr., London
6. Kluwer Fotheringham A S, O"Kelly M E., (1998), Spatial Interaction Models: Formulations and Applications.
7. Paul Longley, Michael Goodchild, David Maguire and David Rhind:, (2005), Geographical Information Systems. Principles, Techniques, Applications and Management. John Wiley & Sons.

Paper III Core Course	Geoinformatics in Agriculture, Soil & Land Evaluation L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 203

Course Objectives: *This course aims to develop understanding of spatial data, geospatial technologies, and Geoinformatics applications in sustainable agriculture, soil health assessment, land evaluation, and ecosystem management for food security and environmental sustainability. This course aligns SDG 2 and SDG 15.*

Learning Outcomes:

After the completion of the course, the students will have the ability to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Recall the basic concepts and terminology of Geoinformatics related to sustainable agriculture, soil resources, and land ecosystem management	**BT1**
CO2	Explain the principles and methods of Geoinformatics for agricultural monitoring, soil assessment, and land evaluation applications	**BT2**
CO3	Apply Geoinformatics techniques to develop sustainable crop management, soil conservation, and land use planning strategies	**BT3**
CO4	Analyze and evaluate the effectiveness of Geoinformatics-based approaches for agricultural productivity, environmental management, and sustainable land resource utilization	**BT4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Estimation & spectral analysis of crops and damage assessment: spectral properties of crops and yield parameters, identification of crops and acreage estimation., vegetation indices, production forecasting through digital analysis, monitoring, condition, and damage assessment, detection of pests and diseases, damages due to droughts and floods, water-logging and salinity, stress detection.	12
Unit 2	Soil classification and mapping: soil types in India, soil survey methods, soil classification, problems with soil identification, mapping of soils using remote sensing and GIS techniques	12
Unit 3	Land evaluation & assessment: land evaluation, role of remote sensing in soil conservation, principle and methods of land assessment, agriculture and soil development, RS & GIS in land evaluation	12
Unit 4	Case Studies: GIS for drawing out action plans & recent development in agro- climatic modelling, watershed planning, remote sensing in agriculture & soil studies	12
	Total	**48**

Text Books:

1. Steven, M.D. and Clark, J.A., 1991, Application of Remote Sensing in Agriculture, Butterworths, London
2. Ghassem Asrar, 1989. Theory and application of optical remote sensing. John Wiley & Sons, New York

Reference books:

1. Space Applications Centre- Manual of procedure for Forest mapping and Damage Detection using satellite data, Report No. IRS-UP/SAC/FMDD/TN/16/90, 1990: pp-58.
2. Space Applications Centre –Status Report on Crop Acreage and Production Estimation, Report No. RSAM/SAC/CAPE/SR/ 25/90, October 1990, pp-253.
3. Brockington, N.R., (1979): "Computer Modelling in Agriculture", Oxford University Press
4. Siva Vandana, (2002), "Sustainable agriculture and food security", Sage Publications New Delhi.

Paper IV Core Course	Geoinformatics in Regional & Urban Planning			Subject Code:
	L-T-P-C: 3-1-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEOI164C204

Course Objectives: *This course aims to develop understanding of geospatial data and Geoinformatics technologies for sustainable regional and urban planning, smart city development, infrastructure management, and community-based spatial decision-making. This course aligns with SDG 11.*

Learning Outcomes:

After the completion of the course, the students will have the ability to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Recall the fundamental concepts and terminology of Geoinformatics in sustainable regional and urban planning	**BT1**
CO2	Explain and analyze different types of spatial data and geospatial technologies used in urban infrastructure and regional planning.	**BT2**
CO3	Apply Geoinformatics tools and techniques to collect, process, analyze, and visualize spatial data for sustainable city and regional development.	**BT3**
CO4	Evaluate and propose improvements in Geoinformatics-based regional and urban planning systems for smart, resilient, and sustainable communities.	**BT4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Basic concept: importance & relevance of Remote Sensing data for urban and regional planning, visual and digital data analysis techniques, scale and resolution concepts, scope and limitations of remote sensing application to urban and regional planning	14
Unit 2	Regional and urban planning: urban and regional mapping, base map preparation, regional, city, intra –city, scale & methodology, urban and regional plan formulation, application of remote sensing techniques in regional plan, master plan	14
Unit 3	Urban analysis: urban analysis, urban growth, trend analysis, change detection, slum development, housing typology and density analysis, population estimation, information system, database organisation- large scale data entry, interpretation manipulation- retrieval- attribute information for urban planning.	10
Unit 4	Case studies: analysis of urban land use change, preparation of master plan in city development, object-oriented gis data modelling for urban design, delineation of socio-infrastructure database into GIS for land use planning	10
	Total	**48**

Text Books:

1. Arnoff, S (1989); Geographical Information Systems: A Management Perspective, WDL Publications, Canada
2. Brench M.C. (1972), City planning and Aerial Information, Harvard University, Cambridge

Reference Books:

1. Burrough, P.A (1988), Principles of Geographical Information Systems for land Resources Assessment, Oxford University Press
2. Subudhi A.P, Sokhi, Roy (2001), Remote Sensing and GIS, Application in Urban and Regional Studies, IIRS, Dehra Dun
3. Subudhi, A.P (1992), Design of Automated Land Use Information System for Town & Country planning, Institute of Town planners, New Delhi.

Paper V Core Course	Remote Sensing and Digital Image Processing and Analysis L-T-P-C: 0-0-8-4 Credit Units: 4 Scheme of Evaluation: (P)	Subject Code: GEOI164C 215

Course Objectives: *This course aims to enhance students' practical knowledge and technical skills in Remote Sensing, Digital Image Processing, and GIS for innovative geospatial applications, infrastructure analysis, spatial planning, and resource management. This course aligns with SDG 9.*

Course Outcomes:
By the end of this course the students will be able to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Define the principles and practical concepts of GIS, Remote Sensing, and digital image processing techniques.	**BT1**
CO2	Classify the nature, characteristics, and sources of geospatial data used in spatial analysis and infrastructure-related applications.	**BT2**
CO3	Develop technical skills in Remote Sensing, digital image processing, and geospatial data analysis using modern GIS tools.	**BT3**
CO4	Apply geospatial concepts and image analysis techniques to study geomorphology, climatology, population distribution, and regional development.	**BT4**
CO5	Evaluate and utilize geospatial technologies for infrastructure planning, natural resource assessment, spatial management, and innovation-driven decision-making.	**BT5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit I	**Lab Work:** Exercise 1 Georeferencing of satellite images based on a georeferenced map Exercise 2 Stacking of images (Image composition) with various spectral bands and generating True, False and Pseudo Colour Composite. Exercise 3 Mosaic of image and clipping the same or any other image by vector polygon boundary	14
Unit 2	Exercise 4 Enhancement using different filtering techniques, Image Fusion Exercise 5 Satellite imagery and feature identification based on interpretation keys	10
Unit 3	Exercise 6 Interpretation of Satellite Imagery in different Bands Exercise 7 Band ratio (NDVI, NDWI, NDSI, NDMI) Exercise 8 Digital Image Classification: Supervised, Unsupervised and accuracy assessment.	12
Unit4	Exercise 9: Interpretation of Thermal Image and Drawing of Isotherms Exercise 10 Generation of slope, profiles from contour digitization from toposheet. Exercise 11 GPS: GPS Survey, Data Import, Processing and Mapping	12

Books Recommended

1. Date, C.J., 1995 : An Introduction to Data Base System, 6th edition, Reading Messachusetts; Adderson Werley.
2. Fraser Taylor, D.R., (ed.), 1980 : Progress in Contemporary Cartography, John Wiley, Chichester U.K.
3. Fraser Taylor, D.R., (ed.), 1983 : Graphic Communication and Design in Contemporary Cartography, John Wiley & Sons Ltd. New York.
4. Jones, C., 1997 : Geographic Information Systems and Computer Cartography, Longman, London.
5. Kraak, M-J., and Ormeling, F., 2004: Cartography: Visualization of Geospatial Data, Pearson Education.
6. Misra, R.P., et al 2014: Fundamentals of Cartography, Concept Publishers, Delhi.

SEMESTER-III

Paper I Core Course	ADVANCED REMOTE SENSING L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 301

Course Objectives: The objective of this paper is to understand the basic concepts of Remote Sensing and to impart to students the skills necessary for remote sensing analysis and interpretation**. The paper's objectives are aligned with the SDG goals 13, and 15.**

Course Outcomes:

After the completion of the course, the students will have the ability to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Gain knowledge of advanced satellites in Remote Sensing, Hyperspectral Remote Sensing, LIDAR Remote Sensing, and their different applications in terrestrial and vegetation mapping.	**BT1**
CO2	Acquire skills in handling instruments, tools, techniques, and modeling while using Remote Sensing technology.	**BT2**
CO3	Get familiarized with various image enhancement and image processing techniques and apply this knowledge to advance the Sustainable Development Goals (SDGs), **Climate resilience, and biodiversity protection**.	**BT3**
CO4	Explore employability opportunities in space organizations.	**BT4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	**Thermal Remote Sensing:** Principles of thermal remote sensing, black body, radiant temperature, radiation from Earth‟s objects, thermal conductivity, thermal capacity, thermal inertia, thermal diffusivity, Thermal Radiometers, scanners, calibration of scanners, mapping with Thermal scanners, Imaging Spectrometer, Application of Thermal Remote Sensing.	14
Unit 2	**Hyper Spectral Remote Sensing** Introduction to Hyperspectral remote sensors/imaging spectrometers, hyperspectral satellite systems, hyper spectral image analysis: atmospheric correction, analysis technique of hyper spectral remote sensing, biophysical modeling, image transmission & compression. Spectroscopy, Image cube, Hyperian/HYSI, Spectral matching, digital spectral data, libraries, application of hyper spectral data, MODIS	12
Unit 3	**Lidar Remote Sensing** Fundamental of LIDAR remote sensing, LIDAR data processing, LIDAR data management and applications (topographic mapping, flood inundation analysis, line-of-sight analysis, forestry, various types of lidar sensors-, vegetation metric calculations, corridor mapping system,), terrestrial and bathymetric laser scanner satellite and its classification. sun synchronous orbit and geostationary orbit, remote sensing satellites in operation: LANDSAT, SPOT, IRS, INSAT, GEOSAT, IKONOS, QUICK BIRD, NOAA, TERRA their sensor characteristics and application.	12
Unit 4	**Microwave & Radar Remote Sensing***:* Concept and principles of Microwave Remote Sensing, SLAR, SAR and Scatterometer, Application of Microwave Remote Sensing. Outlines of Radar Image Interpretations. Image Interpretation: visual and digital interpretation techniques - basic concepts of visual interpretation, tone, color, texture, pattern, shape and contextual features.	10
	Total	**48**

Text Books:

3. Sabins, Floyd F. (2007), Remote Sensing: Principles and Interpretation, H. Freeman and C., New York.
4. Thomas M. Lillesand & Kiefer, Ralph W. (2007), Remote Sensing and Image Interpretation, John Wiley & Sons, New York.

Reference books:

7. Jensen, JR. (2006), Remote Sensing of the Environment- An Earth Resources Perspective, Prentice Hall Inc.
8. Rencz, Andrew N., (1999), Remote Sensing for the Earth Sciences: Manual of Remote Sensing, 3rd ed., John Wiley & Sons, Inc., New York.
9. Curran, P., (1985), Principles of Remote Sensing, Longman, London.
10. Campbell, James B., (2006), Introductory Remote Sensing: Principles and Concepts, Routledge.
11. Gibson, P.J., (2000), Introduction to Remote Sensing, 2nd ed., Taylor & Francis, London.
12. Cracknell, A.P. & Hayes, L.W B., (2007), Introduction to Remote Sensing, Taylor & Francis, London.
13. Schowengerd, R.A. 1995 Techniques for Image processing and classification in Remote Sensing, Academic Press. New York.
14. Siegel, B.S. and Gillespie, A.R. 1994, (eds). Remote sensing and Image Interpretations, John Wiley and Sons, New York.

Paper II Core Course	RESEARCH METHODOLOGY L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 302

Course Objectives:

The course aims to make the students understand the basics of qualitative and quantitative research, literature review, data collection, identification of research problem, formulate research objectives and research questions, formulation of hypothesis and testing, framing of questionnaires, techniques of collection of both qualitative and quantitative data and their analysis. The paper's objectives are aligned with SDG 4.

Course Outcomes:

After the completion of course, the students will have ability to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Define the concepts and tools of research.	**BT1**
CO2	Infer ideas that can be taken up for research work through literature review.	**BT2**
CO3	Develop hypothesis and research questions.	**BT3**
CO4	Identify appropriate data collection and sampling techniques.	**BT4**
CO5	Interpret the various types of data along with critical evaluation.	**BT5**
CO6	Design and develop a scientific research report.	**BT6**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Research: definition, types (pure and applied) classification, literature review. research methodology in geosciences; defining a research problem; statement of the problem; objectives, and hypothesis/ research questions, database and methodology, significance, review of research works and bibliography and references.	14
Unit 2	Routes of explanation: inductive and deductive, hypothesis, theories, laws and models, research question, objectives and significance of research, research design: data collection and analysis	12
Unit 3	Presentation of research findings: types, role and significance; questionnaire design (open, closed, structured, and non-structured), data collection, post field processes: construction of data matrix, data processing and analysis; role of quantitative techniques in geography. report writing and presentation, scientific journals (impact factor, citation), introduction to Web of Science, SCOPUS, Mendeley and Google scholar	12
Unit 4	Ethics in Scientific Research: Plagiarism- classification and prevention; Intellectual property rights; Research report writing: Structural components and presentation. Preliminary idea about URKUND, TURNITIN, DRILLBIT	10
	Total	**48**

References

Text Books:

1. Harvey, D, 1969: Explanation in Geography, Scientific Publisher, Jodhpur.
2. Lenon, B., Cleves, P. 2015. Geography Fieldwork and Skills, Harper-Collins.

Reference Books:

3. Evans, M., (1988): "Participant Observation: The Researcher as Research Tool" in Qualitative Methods in Human Geography, eds. J. Eyles and D. Smith, Polity.
4. Special Issue on "Doing Fieldwork" The Geographical Review 91:1-2 (2001).
5. Stoddard, R. H., (1982): Field Techniques and Research Methods in Geography, Kendall/Hunt.
6. Wolcott, H., (1995): The Art of Fieldwork, Alta Mira Press, Walnut Creek, CA.
8. Northey, N., Draper, D., Knight, D.B. 2015. Making Sense in Geography and Environmental Sciences:A Student's Guide to Research and Writing, 6th ed, Oxford University Press.
9. Parsons, T., Knight, P.G. 2015. How To Do Your Dissertation in Geography and Related Disciplines,3rd ed, Routledge.

Paper III Core Course	Techniques in Advanced Remote Sensing and Spatial Modelling L-T-P-C: 0-0-8-4 Credit Units: 4 Scheme of Evaluation: (P)	Subject Code: GEOI164C 313

Course Objectives: The course aims at increasing the practical knowledge of the students in the field of GIS and Remote sensing and its application in Geographical studies. The paper's objectives are aligned with SDG 9.

Course Outcomes:
By the end of this course the students will be able to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Define the principles and concepts involved in GIS and Remote Sensing practical.	**BT1**
CO2	Classify the nature, characteristics, and sources of geospatial data.	**BT2**
CO3	Develop the skills and technical capabilities of the students.	**BT3**
CO4	Simplify the application of concepts related to Geomorphology, Climatology, and Population Geography. Applied learning as a catalyst to advance SDG 9 by fostering scientific research, industrial innovation, and sustainable infrastructure development.	**BT4**
CO5	Inspect geospatial tools and technologies to create and analyse geospatial data for natural resource assessments, planning, and management-related applications.	**BT5**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	**Lab Work:** Exercise 1 Image manipulation / enhancements and interpretation and feature identification Exercise 2 Classification- Supervised and Unsupervised Exercise 3 Unsupervised classification and recoding of enhanced image and derivation of statistics	14
Unit 2	Exercise 4 Preparation of various thematic maps: Drainage – TIN – DEM – slope - aspect – land use/ land cover Exercise 5 Mapping of temporal changes in river courses using Remote sensing and GIS techniques Exercise 6 Spatial and Non-Spatial Data: Spatial: linking features attributes, ways to view data, Metadata non-spatial understanding tables, field types, table manipulations, table relationship, joins, relates, creation of graphs and reports	12
Unit 3	Exercise 7: Preparation of soil map, - land capability assessment (using Analytical Hierarchy Process) - soil erosion estimation by using RUSLE/USLE model Exercise 8: Vegetation cover mapping from satellite images, Computation of NDVI, SAVI, NDWI, NDBI from 1C/1D/ ResourceSat/ LANDSAT data/ SAR data	12

	Exercise 9: Temporal Land use / Land cover studies, change detection analysis and accuracy assessment Exercise 10: Digital classification for forest cover mapping and change detection studies, estimation of above ground biomass and Carbon Stock	
Unit 4	Exercise 11: Time-series data (climatic attributes, river discharge, forest cover etc) analysis by using statistical techniques Exercise 13: Network Analysis - shortest path – best path – service area – OD cost matrix - Location and Allocation - route tracing – proximity analysis – site suitability –address matching – (using ArcGIS software) Exercise 12: Introduction to Differential GPS (DGPS): Principle and Function. Use of DGPS in Topographical Survey. Exercise 13: Introduction to Drone Survey and Mapping	10
	Total	**48**

Paper IV Core Course	Geoinformatics in Disaster Management			Subject Code:
	L-T-P-C: 4-0-0-4	Credit Units: 4	Scheme of Evaluation: (T)	GEOI164C304

Course Objectives: The objective of the course is to provide the students with an understanding about the fundamental concept of hazards and disasters and usage of Geoinformatics in its mitigation. The paper's objectives are aligned with the SDG goals 3, 6, and 13.

Learning Outcomes:
After the completion of the course, the students will have the ability to:

CO1 Define hazards and disasters, their characteristics and the role of Geoinformatics in its mitigation.
CO2. Infer basic understanding of different hazards.
CO3. Apply GIS for modelling management of various hazards and develop skills to tackle environmental and Climate resilience.
CO4. Analyze case studies to understand the hazards prevalent in India and focus on its mitigation.

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Introduction: Hazards and disasters, their types, and characterization, Zonation of hazards, natural and human induced disasters, Disaster and National losses, historical perspective of disasters in India, Fundamental concept of Disaster Management, Government, NGOs and peoples participation disaster management, Existing organization structure for managing disasters in India, Geoinformatics in disaster mitigation.	14
Unit 2	Hazards: Landslide, Earthquake, Mining hazards (Land subsidence, Mine flooding etc.), Volcanic hazards, Groundwater hazards, Glacial hazards, Flash floods, River floods, Dam burst, Cloud burst, Cyclones, Coastal hazards and Drought, Forest hazards (Deforestation, Degradation and Forest fire), Land & soil degradation, Desertification, Pollution (Water, air, and soil)	14
Unit 3	Geoinformatics Applications: Geoinformatics models in managing forest fires, floods, landslides, cyclone and earthquake mapping.	10
Unit 4	Case Studies: Earthquakes in India, Floods in Indo Gangetic plains, Landslides in Himalayan region, Drought in Indian plateau regions.	10
	Total	**48**

Text Books:

1. P.S. Roy (2000) Natural Disaster and their mitigation. Published by Indian Institute of Remote Sensing.

Reference Books:

1. Sdidmore A (2002) Environmental Modeling with GIS & Remote Sensing, Taylor & Francis.
2. Anji Reddy. M. (2004) Geoinformatics for Environmental Management. B. S. Publication.

Paper V Core Course	GEO-INFORMATICS IN FORESTRY L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C305

Course Objectives:

- To study the Spectral characteristics of Vegetation
- To study the integrated analysis of GIS in forest management and **SDG 15**.

Course Outcomes:

After the completion of the course, the students will have the ability to:

CO 1: The outcome of this subject is to know how to prepare Biomass estimation, forest fire map etc. The learning outcome can be applied in forest conservation and biodiversity protection.

CO 2: To prepare different models using software **to estimate forestry and biodiversity-related issues**.

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Forest: Introduction (meaning and its role in environmental protection) and Global distribution of forest cover and its change. Forest types of India. Forestry: introduction and concept of forestry. Role of RS and GIS in forestry	14
Unit 2	Interaction of EMR with vegetation and spectral characteristics of vegetation. Temporal characteristics of vegetation. Vegetation indices. Forest cover mapping through RS and GIS	12
Unit 3	Forest types and forest density mapping. Remote Sensing application in forest covers change detection. Remote Sensing application in mapping of stressed vegetation. Study of association between rock and forest types using RS and GIS. Bio diversity studies using RS and GIS. Wildlife habitat analysis using RS and GIS	12
Unit 4	Role of microwave remote sensing in forest studies. Biomass estimation by non-destructive method. Growing stock estimation using RS and GIS. Remote sensing application in formulation of forest working plan. Biological invasion and monitoring of invasive species through RS and GIS. Forest management information system (FMIS)	10
	Total	**48**

Books Recommended

1. Anji Reddy, M. 2004: Geoinformatics for Environmental Management. B.S. Publications
2. Franklin S.E. 2001. Remote Sensing for Sustainable Forest Management. Lewis Publication
3. Gupta, R.P.., 1990: Remote Sensing Geology. Springer Verlag.
4. Jensen, J.R. 2000: Remote Sensing of the Environment: An Earth Resource Perspective. Prentice Hall
5. Lillesand, T.M., and Kieffer, R.M., 1987: Remote Sensing and Image Interpretation, John Wiley.

REFERENCES

1. Steven. M.D. and Clark. J.A, "Applications of Remote Sensing in Agriculture", Butterworths, London 1990.
2. Remote Sensing Applications Group", Space Applications Centre, Crop Average and production Estimation (CAPE): An Anthology from January 1986 - June 1996. (Publications in Journals, Seminars I Symposium proceedings), Ahmedabad, August 1996.
3. Negi. S.S," A Handbook of forestry. International Book distributors", Dehradun, 1986. Space Applications Centre, Manual of procedure for Forest mapping and Damage Detection using satellite data, Ahmedabad, 1990.

Core Course	FIELD-BASED RESEARCH I Credit Units: 8 Scheme of Evaluation: (P)	Subject Code: GEOI164C 324

Course Objectives: To develop students' ability to identify geographical research problems, design field-based studies, apply appropriate research methodologies, and prepare and present scientific research proposals.

Course outcome:
CO1: Identify and formulate geographical research problems.
CO2: Conduct literature review and develop research designs.
CO3: Design field surveys and appropriate methodologies for data collection.
CO4: Prepare and present a scientific research proposal.

Course Outcomes:
After successful completion of the course, the students will be able to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Recall the basic principles of research design, methods, and ethics.	**BT1**
CO2	Interpret and analyze different types of research methods and data collection techniques.	**BT2**
CO3	Utilize research-based approaches to develop a hypothesis or research question and design a research project.	**BT3**
CO4	Analyze and evaluate the accuracy, reliability, and limitations of data collected and used in the research project.	**BT4**
CO5	Critique and propose improvements to research design and methods.	**BT5**
CO6	Design and develop a research project proposal, including research question/hypothesis, methods, data collection, and analysis techniques.	**BT6**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Identification of Research Problem and Study Area Identification and selection of a geographical research problem based on field observations, internship experiences, or local/regional issues. The study may focus on: a. Agriculture, Industry and Mining b. Environmental Problems, Disasters and Hazards c. Natural Resource Assessment and Management d. Climate Change and Sustainability e. Rural and Urban Development f. Watershed, Basin and Terrain Studies Selection and delineation of study area, statement of the problem, significance and scope of the study.	12
Unit 2	Review of Literature and Research Design • Review of relevant literature • Identification of research gaps • Formulation of aims and objectives • Development of research questions and hypotheses • Conceptual and theoretical framework • Preparation of work plan and research design	12
Unit 3	Field Survey Design and Methodology • Selection of data sources (Primary and Secondary) • Sampling techniques and survey design • Preparation of questionnaire, schedule and interview formats • GPS-based field data collection techniques • Observation and Participatory Rural Appraisal (PRA) methods • Pilot survey and validation of research tools • Ethical considerations in field research • Field study will be commence in the last week of October	12
Unit 4	Research Proposal Preparation and Presentation Preparation of research proposal including: • Title of the Study • Introduction and Background • Statement of the Problem • Objectives and Research Questions • Study Area Description • Database and Methodology • Literature Review • Expected Outcomes • Time Schedule and Work Plan Presentation of the proposal through PowerPoint and discussion before the departmental committee.	12
	Total	**48**

Learning Outcomes:

Note: Submission of project report in prescribed format and on specified date is mandatory. Equal weightages of marks for each stage of the work (upto 3rd stage) for internal evaluation of the project by the supervisor (60% of end semester examination). 40% of the total marks of end semester examination is for viva-voce and final presentation to be evaluated by an external examiner.

Text Books:

As per the list of given in syllabus based on topic selected

Reference Books:

As per the list of given in syllabus based on topic selected

SEMESTER-IV

Paper I Core Course	GEOINFORMATICS IN GEOMORPHOLOGICAL STUDIES L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 401

Course Objectives: This course intends to show the rationale behind the use of remotely sensed data and its advantages and disadvantages and illustrate how GIS/GPS methodologies can be used to address spatial analysis from the theoretical and practical perspective. This paper is aligned with SDGs 13 & 15.

Course Outcomes:

After successful completion of the course, the students will be able to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Define the basic concepts of application of geoinformatics in geomorphological science.	**BT1**
CO2	Interpret geomorphological factors through aerial photography and satellite imagery.	**BT2**
CO3	Utilize geostatistical analysis techniques for geomorphological mapping, support natural habitat preservation, enhance biodiversity conservation, and strengthen hazard management.	**BT3**
CO4	Apply this knowledge for geomorphic mapping and analysis.	**BT4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	**Concepts of Modern Geomorphology**: Geomorphology and its applications in Natural resources inventory. Geomorphology and its applications to Geoinformatics	12
Unit 2	**Geomorphic Environments:** The Fluvial Systems. Coastal and Marine geomorphology. Aeolian, Glacial, Karst and Dune Environments. M.O. Ridges, Bathymetry.	12
Unit 3	**GIS in Geo-environment Management** Impact of slope, badlands, pediments, streams in geomorphic evolution. Geomorphic controls on the groundwater resources of Coastal, Island and hinterland terrains. Geomorphological factors for Pollution management	12
Unit 4	**Hazard Analysis**. Application of Remote Sensing and GIS in quantitative and quantitative interpretations of „risk area mapping" including floods, earthquakes and Tsunami affects terrains. use of digital data products in assessing damage due to earthquakes, Landslide, Flood. study of mining hazards.	12
	Total	**48**

Reference Books:

1. Fundamentals of Photogeology, Geomorphology – Verstappen – TTC Holland.
2. Thornbury,W. D., 2004, Principles of Geomorphology, CBS Publ., 5-570.
3. Wathern, P 1988, EIA: Theory & Practice. Unwin Hyman, London, 1-17.
4. Wood, C. 1995 EIA: A Comparative Review. Longman. 87-255.
5. Pethick, J. 1984. An introduction to Coastal Geomorphology, Edward Arnold, London, 259p.
6. Ritter, D.F., R.C. Kochel and J.R. Miller (2011) *Process Geomorphology, 5th edition*. McGraw Hill, NY.
7. Summerfield, M.A. (Editor), 1991. Global Geomorphology: An introduction to the study of landforms, John Wiley and Sons Ltd., New York: 560p.
8. Thornbury, W.D. (1969): Principles of Geomorphology, Wiley Eastern Limited, New Delhi: 594 p.
9. Tinkler, 1985. A short history of Geomorphology, Croom-Helm, London.
10. Rice (1998): Fundamentals of Geomorphology.
11. Kale & Gupta (2001): Introduction to Geomorphology.

Paper II Core Course	GEOINFORMATICS IN WATER RESOURCES L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C402

Course Objectives: *This course aims to make the students understand the basic concepts and principles of geoinformatics in the context of water resources management. This paper addresses critical issues of water under SDG 6.*

Learning Outcomes:

After the completion of the course, the students will have the ability to:

Recall the basic concepts and terminology related to Geoinformatics and water resource management.

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Recall the basic concepts and terminology related to Geoinformatics and water resource management.	**BT1**
CO2	Interpret maps and other visual representations of water resources data.	**BT2**
CO3	Utilise Geoinformatics tools and techniques to collect, analyse, and visualise water resource data and perform analytical techniques to quantify and manage the water-related issues in different parts of the country.	**BT3**
CO4	Evaluate the accuracy and reliability of Geoinformatics-derived water resource data.	**BT4**

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Basic Concept: Hydrologic Cycle, hydrological parameters, Watershed characterization, delineation and codification, Watershed problems and management strategy, Geoinformatics approach for watershed prioritization, drainage morphometric analysis	14
Unit 2	Remote Sensing in surface-subsurface water exploration: application of remote sensing in hydro geomorphological interpretation for ground water exploration, water quality monitoring through remote sensing, geophysical methods for groundwater exploration.	14
Unit 3	Applications in Water Resources: flood prediction, drought evaluation, snow cover mapping, reservoir sedimentation evaluation, geoinformatics based runoff & hydrological modelling, flood hazards modelling, snowmelt runoff modelling.	10
Unit 4	Case Studies: hydro geomorphological mapping in Plateau Region, flood prone zone mapping in Indo Gangetic Plains, water harvesting initiatives in urban built up lands, drought assessment in Jharkhand.	10
	Total	**48**

Text Books:

Schultz, G. A. and Engman, E. T., (2000), Remote Sensing in Hydrology and Water Management, Springer-Verlag, Berlin, Germany.

Reference Books:

1. Murthy, J. V. S. (1994). Watershed Management in India. Wiley Eastern Ltd., New Delhi.
2. Todd David Keith., (2005), Groundwater Hydrology, John Wiley & Sons, New York, Second Edition.
3. Schultz, G.A. & Engman, E.T., (2000), Remote Sensing in hydrology and water management, Springer-Verlang, Berlin, Germany.

Paper III Core Course	GEOINFORMATICS IN CLIMATE CHANGE STUDIES L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 403

Course Objectives: *Climate change and its corollary global warming are the much talked-about these days for there is an impending danger to the earth we live in by the climate change caused primarily by the human activities on the earth. Climate change has already brought untold sufferings to the world that the world countries met several times to work towards a strategy for reducing global warming and the consequent climate change. This paper offers deep insights into the working of climate change and how to overcome it.* ***This paper focuses on global geoscience issues by addressing SDG 13.***

Course Outcomes:

After successful completion of the course, the students will be able to:

CO1. Define the basic concepts of earth‟s atmosphere

CO2. Interpret causes effects and importance of climate change.

CO3. Utilise geostatistical analysis techniques for all the climatic attributes mapping and gain technical capability of analyzing climate data and formalizing adaptive management.

CO4. **Apply** this knowledge for global climate change issues

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	**Earth System Dynamics**: Introduction to atmosphere, hydrosphere, biosphere, lithosphere, and human interventions in earth system dynamics and operations, anthropogenic activities and global warming.	6
Unit 2	**Climate Change: Concept and Process**: Introduction, Concept, causes, effects, measures, importance of climate change, climate change and energy, climate change and emerging diseases, climate and change and community.	6
Unit 3	**Issues in Climate Change**: Global warming, greenhouse effect, carbon cycle, nitrogen cycle, water cycle, ozone depletion, floods, droughts and weather variations, El-NINO and La-NINA, changing ecosystems, snow / glaciers melting, sea level changes	12
Unit 4	**Geoinformatics Applications**: Hazards, risks and vulnerability analysis relating to global warming, floods and droughts, and weather variations, ecosystems changes, and snow/glaciers melting, energy studies, health and diseases studies and other case studies.	10
	Total	**48**

References

1. Climate Change: A Multidisciplinary Approach- Burroughs, W.J.
2. The Suicidal Planet: How to Prevent Global Climate Change- Mayer Hillman,
3. Field Notes from a Catastrophe: Man, Nature, and Climate Change- Kolbert, Elizabeth.
4. Cradle to Cradle: Remaking the way we make things William McDonough,
5. Integration of GIS, remote sensing, Photogrammetry and cartography: the Geoinformatics approach - Ehlers, M.

Paper IV Core Course	GEOINFORMATICS IN LAND RESOURCE MANAGEMENT L-T-P-C: 3-1-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 404

Course Objectives:

- *To study the landuse and management system of rural and urban.*
- *To study the integrated analysis of GIS in landuse planning*
- *To understand the concepts and principles and use the tools and techniques of GIS for efficient planning and management of urban area. This paper addressed land and soil resource management under SDG 12 & 15.*

Course Outcomes:

After the completion of the course, the students will have the ability to:

CO1. List the basic concepts of land utilization in rural as well as urban CO2. Surveying land-soil-water resources in a proposed area, and gain RSGIS technology to preserve the land and soil resources for sustainable development in an ever-shrinking soil and land area.

CO3. Planning of urban land use specifically in India

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	**Land Use**: Land use systems, land utilization types; land use classifications – rural and urban land uses and land use patterns, Municipal Lands and Open Spaces in Cities and Town, Agriculture and Forest Land Management, Recreational Lands, Wetland Management. .	14
Unit 2	**Data Sources for Land Evaluation**: Land-soil-water resources surveys; remote sensing and GPS surveys of land uses; land use /land cover classification from remotely sensed data; vegetation indices, supervised and unsupervised classification.	12
Unit 3	**Land Evaluation**: The logical basis of land evaluation; land evaluation for land use planning; Biophysical models of land evaluation, the FAO two-stage approach to land evaluation; other approaches to land capability and suitability classifications	12
Unit 4	**Land Use Planning**: The importance and difficulty of land use planning, Urban Land Use Planning Strategies, land use policies, principles of land use planning and land use management; urban land use planning, critical issues of land use planning in India.	10
	Total	**48**

References

1. Action Planning for Cities: A Guide to Community Practice - Hamdi, Nabeel
2. Applied Remote Sensing for Urban Planning, Governance and Sustainability – Netzband Maik
3. Remote Sensing of Urban and Suburban Areas - Tarek Rashed, Carsten Jürgens
4. Remote sensing and urban analysis - Jean-Paul Donnay, Michael John Barnsley
5. Urban Remote Sensing - Qihao Weng, Dale A. Quattrochi
6. Radar Remote Sensing of Urban Areas, Remote Sensing and Digital Image Processing - Soergel Uwe
6. Analysis of Urban Growth and Sprawl from Remote Sensing Data - Basudeb Bhatta

Paper IV Core Course	GEO INFORMATICS IN ENVIRONMENTAL MANAGEMENT L-T-P-C: 4-0-0-4 Credit Units: 4 Scheme of Evaluation: (T)	Subject Code: GEOI164C 405

Course Objectives: *This course intends to show the rationale behind the use of remotely sensed data in the study of environmental management and illustrate how GIS/GPS methodologies can be used to monitor environmental problems from the theoretical and practical perspective. This paper serves the objectives of SDGs 12 & 13.*

Course Outcomes:

After successful completion of the course, the students will be able to:

CO1. Monitoring and mapping ecological and biological aspects of environment.

CO2. Utilize geostatistical analysis techniques for water quality mapping.

CO3. Apply this knowledge for Environmental management plan Detailed Syllabus:

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Biotic and Abiotic environment, Environmental pollution and types of environmental pollution. Change detection studies with the help of multi temporal data. Remote sensing in pollution monitoring; Concept of environmental management.	6
Unit 2	Water quality analysis (based on different parameters); Remote sensing in water quality mapping monitoring and management. Sewage management – introduction, classification and environmental problems.	6
Unit 3	Anthropogenic disasters: introduction and types. Application of remote sensing & GIS in management of man-made disasters., power plants, nuclear waste management, global and Indian scenario.	12
Unit 4	Impact assessment – basic concepts, environmental impact assessment (EIA) methods. Environmental analysis and environmental monitoring for sustainable development through RS & GIS. EIA of mining areas and river valley project through remote sensing. Environmental management plan (EMP), its importance and role of GIS in preparation of EMP	10
	Total	**48**

References

1. Allah Brimicomber, GIS Environmental Modeling and Engineering, Taylor and Francis, 2003
2. Savigny D De and Wijeyaratne. P.GIS for Health and Environment, Stylus publication, 1994.
3. Paul A Longley, Michael F Goodchild, David J Maguire, David W Rhind, Geographical Information Systems, Volume I and II,John Wiley and Sons, Inc., 1999.
4. Van Dijk M.G.Bos, GIS and Remote Sensing Techniques in Land-And-Water Management, Kluwer Academic Publishers, 2001.
5. Juliana Maantay, John Ziegler and John Pickles, GIS for the Urban Environment, ESRI Press, 2006.
6. Ahmad, Y. J and Sammy, G. K 1985 Guidelines to Environmental Impact Assessment in Developing Countries. Hodder & Stoughten, London. 26-82.
7. Anonymous, 1992. Overseas Development Administration-manual of Environmental Appraisal.ODA, London-II Edition. 8-16.
8. Anonymous, 1993. NATO-Methodology, Evolution and Scope of EIA, Report 197, NATO Brassiles, 3-12.
9. Beanlands G. E. &Dunniker, P. N 1984 An Ecological Frame work for Environmental Impact Assessment, Journal of Environmental management. 18:267-277.

10. Meenakshi, P., 2006, Elements of Environmental Science and Engineering. Printice Hall. 2-307.
11. Murthy, K. S. 1988. National Environmental Policy Act (NEPA) Process. CRC Press, Boca Raton USA, 1-18.
12. Ortolano, L. 1993. Control on Project Proponents and EIA Effectiveness. The Environmental Professional, Vol. 15:350-363.

Paper II Core Course	FIELD-BASED RESEARCH II Credit Units: 12 Scheme of Evaluation: (P)	Subject Code: GEOI164C 423

Course Objectives: *The course aims to enable students to apply GIS to real-world problems, using data from a range of sources, including remote sensing, GPS, and survey data.*

Course outcome:
CO1: Conduct independent geographical research through systematic field investigation.
CO2: Apply appropriate field survey, GIS, Remote Sensing, and statistical techniques for geographical analysis.
CO3: Process, analyse and interpret primary and secondary spatial data.
CO4: Prepare scientific research reports and thematic maps following academic standards.
CO5: Present and defend research findings effectively through oral presentations and viva-voce examinations.

Course Outcomes:
After successful completion of the course, the students will be able to:

Course Outcome (CO)	Course Outcome Description	Bloom's Taxonomy
CO1	Recall the basic principles of research design, methods, and ethics.	**BT1**
CO2	Interpret and analyze different types of research methods and data collection techniques.	**BT2**
CO3	Utilize research-based approaches to develop a hypothesis or research question and design a research project.	**BT3**
CO4	Analyze and evaluate the accuracy, reliability, and limitations of data collected and used in the research project.	**BT4**
CO5	Critique and propose improvements to research design and methods.	**BT5**
CO6	Design and develop a research project proposal, including research question/hypothesis, methods, data collection, and analysis techniques.	**BT6**

Learning Outcomes:

weightages of marks for each stage of the work (upto 3rd stage) for internal evaluation of the project by the supervisor (60% of end semester examination). 40% of the total marks of end semester examination is for viva-voce and final presentation to be evaluated by an external examiner.

Text Books:
As per the list of given in syllabus based on topic selected

Reference Books:
As per the list of given in syllabus based on topic sel

Detailed Syllabus:

Modules	Topics and Course Content	Periods
Unit 1	Field Data Collection and Database Development Execution of field survey and collection of primary and secondary data •Household and Community Surveys • GPS Survey and Ground Truthing • Environmental and Socio-economic Data Collection • Data Coding, Classification and Database Preparation • Compilation of Secondary Data	12
Unit 2	Data Processing, Mapping and Analysis • Data Tabulation and Statistical Analysis • GIS Database Creation • Preparation of Thematic Maps • Graphs, Charts and Diagrams • Spatial and Temporal Analysis • Interpretation of Survey Findings Progress presentation of fieldwork and preliminary analysis.	12
Unit 3	Project Report Writing Preparation of dissertation/report in prescribed format: • Introduction • Review of Literature • Study Area Profile • Methodology • Results and Analysis • Discussion • Conclusions and Recommendations • References and Appendices	12
Unit 4	Final Presentation and Viva-Voce • Submission of Final Project Report • Preparation of PowerPoint Presentation • Research Communication and Presentation Skills • Final Project Presentation • Viva-Voce Examination	12
	Total	**48**

Note:

*** All the field survey will be commence in the month of October**

*****Submission of project report in prescribed format and on specified date is mandatory. Equal weightages of marks for each stage of the work (upto 3rd stage) for internal evaluation of the project by the supervisor (60% of end semester examination). 40% of the total marks of end semester examination is for viva-voce and final presentation to be evaluated by an external examiner.

Text Books:

As per the list of given in syllabus based on topic selected

Reference Books:

As per the list of given in syllabus based on topic selected